ELECTRONICS AND COMMUNICATION ENGINEERING

COMMUNICATION SYSTEM



Baseband Transmission

Lecture No.: 2

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1. Types of Sampling



1. Practice Questions
2. NAT Questions

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Nyquist Sampling

Example 1

Find the sampling frequency of the signal $m(t) = \sin 20\pi t = \sin \omega_m t$.

minimum
or
Nyquist rate

$$f_m = \frac{\omega_m}{2\pi} = \frac{20\pi}{2\pi}$$

$$f_m = 10\ Hz$$

$$f_s = 2f_m = 20\ Hz.$$

Nyquist Sampling

$AT\,\text{sinc}(tT) \xleftrightarrow{FT} A\,\text{rect}\left(\frac{f}{T}\right)$

Coeff of sinc $\longrightarrow$ width of rect

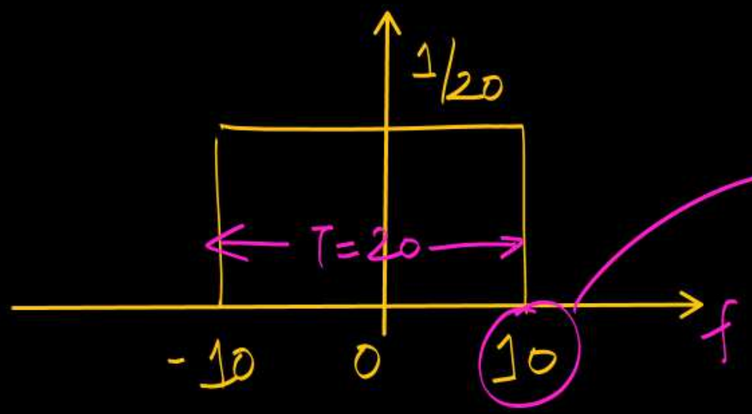
Example 2

Find the sampling frequency of the signal m(t) = sinc 20t

$\text{Sinc}(20t) \xleftrightarrow{FT} \frac{1}{20}\,\text{rect}\left(\frac{f}{20}\right)$

$T = 20, \quad AT = 1$

$A = \frac{1}{T} = \frac{1}{20}$



$f_m = 10\,Hz$

$f_s = 2f_m = 20\,Hz$ ✓

$\text{sinc}(20t) = \frac{\sin(20\pi t)}{20\pi t}$

$\omega_m = 20\pi$

$f_m = \frac{20\pi}{2\pi} = 10\,Hz$

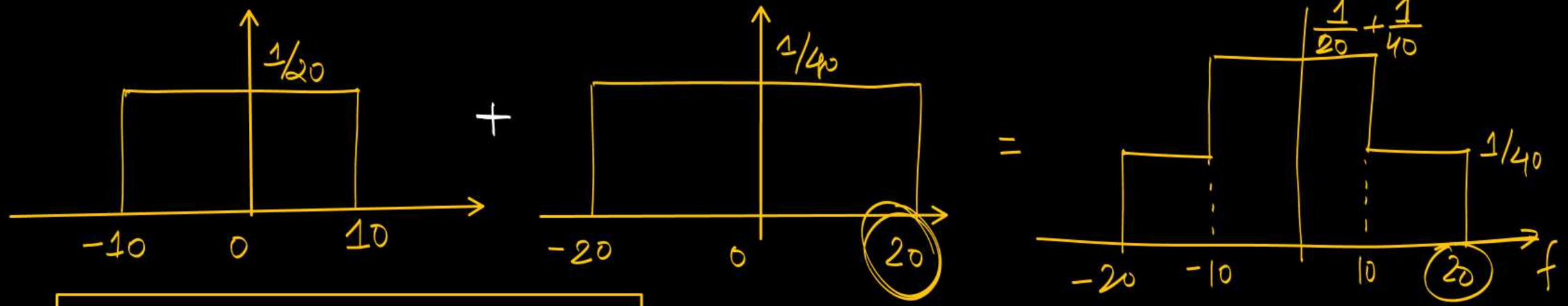
$f_s = 2f_m = 20\,Hz$ ✓

Nyquist Sampling

m(t) → has two freq. component that is why it is called multitone.

Example 3

Find the sampling frequency of the signal $m(t) = \underbrace{\text{sinc } 20t}_{f_1} + \underbrace{\text{sinc } 40t}_{f_2}$



$$(f_s)_{min} = 2 \times \max(f_1, f_2 \cdots f_n)$$

$f_m = 20\text{Hz}$

$f_s = 2f_m = 40\text{Hz}$.

Nyquist Sampling

$m(t) = m_1(t) \cdot m_2(t)$.

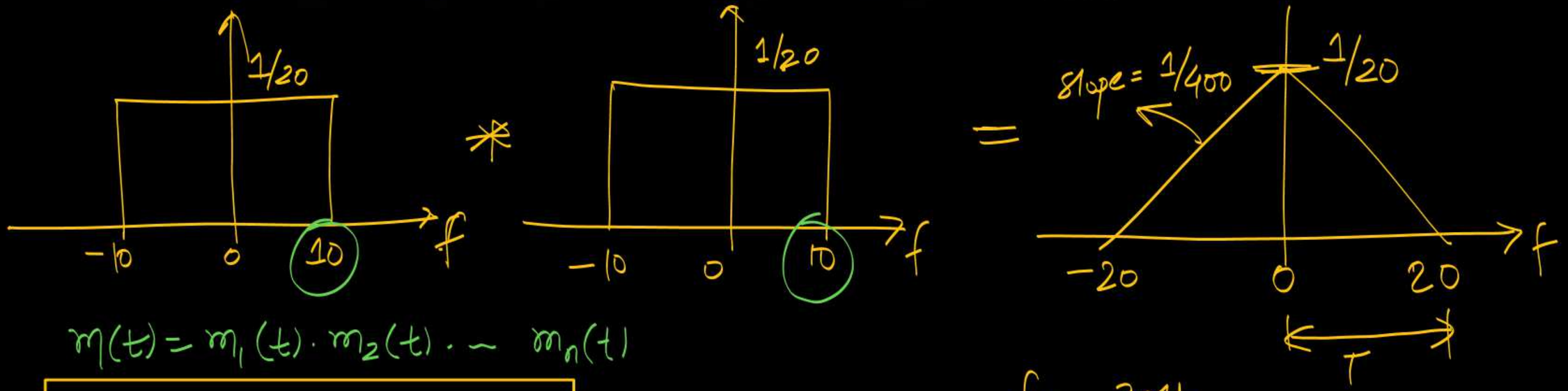
$A\,\text{sinc}^2(tT) \Longrightarrow A\,\text{tri}\left(\frac{f}{T}\right)$

T = Half width

$AT = 1 \Rightarrow A = 1/20$

Example 4

Find the sampling frequency of the signal $m(t) = \text{sinc}^2\,20t$ $= \text{sinc}(20t) \cdot \text{sinc}(20t)$



$m(t) = m_1(t) \cdot m_2(t) \cdot \sim m_n(t)$

$$f_s = 2(f_1 + f_2 + \ldots + f_n)$$

$f_m = 20\,Hz$

$f_s = 2f_m = 40\,Hz$.



Nyquist Sampling

Example 5

Find the sampling frequency of the signal $m(t) = \underbrace{\text{sinc}^3(20t)}_{f_1=30\text{Hz}} . \underbrace{\text{sinc}^2(40t)}_{f_2=40\text{Hz}} . \underbrace{\text{sinc}(10t)}_{f_3=5\text{Hz}}$

$$f_s = 2(f_1 + f_2 + f_3)$$
$$= 2(30+40+5)$$
$$= 150\text{Hz}$$

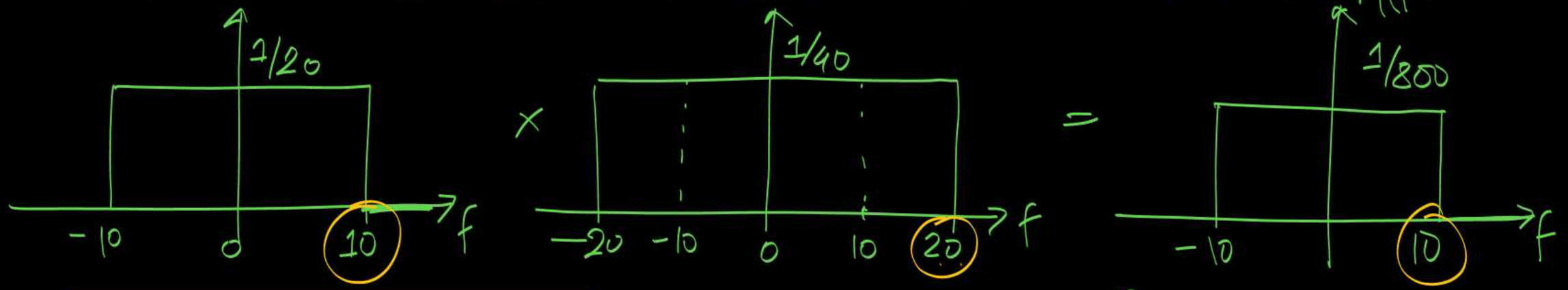
Nyquist Sampling

$m(t) = m_1(t) * m_2(t) * \dots * m_n(f)$

f_1 f_2 f_n

Example 6

Find the sampling frequency of the signal m(t) = sinc 20t * sinc 40t



$f_m = 10\,Hz$

$f_s = 2f_m = 20\,Hz$

$$f_s = 2 \times \min(f_1, f_2, \dots f_n)$$



Nyquist Sampling

Example 7

Find the sampling frequency of the signal $m(t) = \text{sinc}^3(60t) * \text{sinc}^2(40t) * \text{sinc}(20t)$

$f_1 = 90\,Hz$ $\quad f_2 = 40\,Hz$ $\quad f_3 = 10\,Hz$

$f_s = 2 \times \min(90, 40, 10)$

$= 2 \times 10$

$= 20\,Hz$.

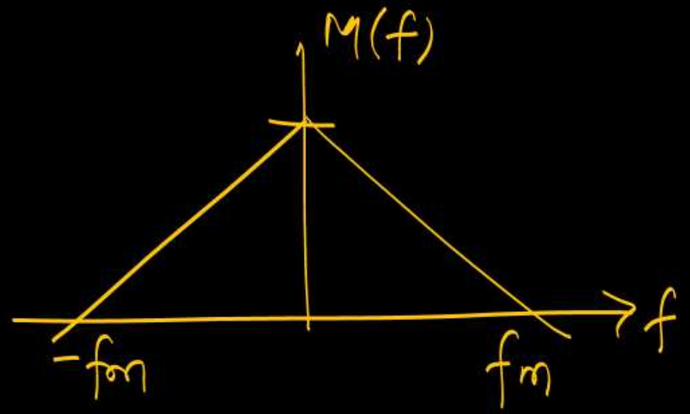
Nyquist Sampling

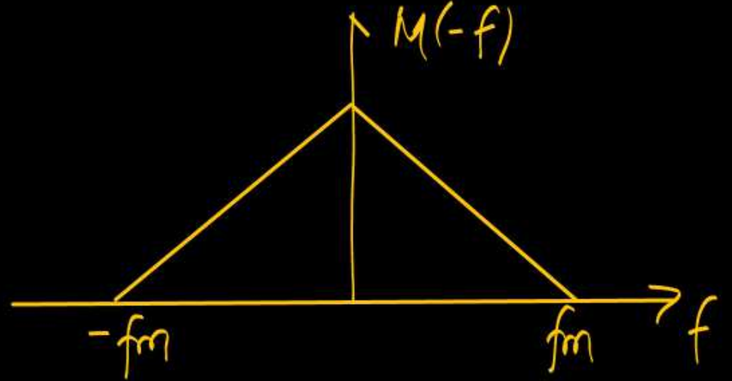
1. Nyquist rate of m(−t)

$$m(t) \xleftrightarrow{FT} M(f)$$

$$m(-t) \xleftrightarrow{FT} M(-f)$$

$$f_s = 2f_m$$





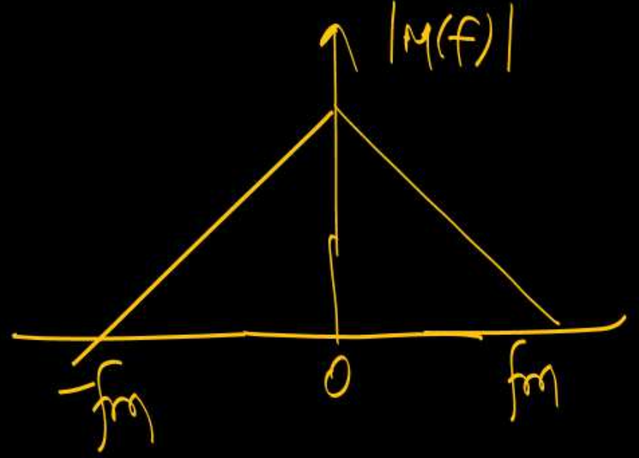


Nyquist Sampling

2. Nyquist rate of $m(t - t_0)$

$$m(t) \longleftrightarrow M(f)$$

$$m(t - t_0) \longleftrightarrow e^{-j2\pi f t_0} M(f)$$



$f_s = 2f_m$



Nyquist Sampling

Signal	Nyquist Rate
$m(t)$	$2f_m$
$m(-t)$	$2f_m$
$m(t \pm t_0)$	$2f_m$
$m(at)$	$(2f_m) \times a$
$m(-at + b)$	$(2f_m) \times a$
$\frac{dm(t)}{dt}$	$2f_m$
$\int_{-\infty}^{t} m(\lambda)d\lambda$	$2f_m$



Nyquist Sampling

The signal $x_1(t)$ is bandlimited to 2 kHz and $x_2(t)$ is bandlimited to 3 kHz. Find the Nyquist rate for the following Signals.

1. $x_1(t) \longrightarrow f_1 = 2\,KHz \quad f_s = 4\,KHz$
2. $x_2(t) \longrightarrow f_2 = 3\,KHz \quad f_s = 6\,KHz$
3. $x_1(t) + x_2(t) \longrightarrow$ multitone $\quad f_s = 2f_2 = 2 \times 3 = 6\,KHz.$



Nyquist Sampling

4. $x_1(t) \times x_2(t)$ multiplication. $f_s = 2(f_1 + f_2) = 2(2+3) = 10$ KHz.

5. $x_1(t) * x_2(t)$ Convolution $f_s = 2 \times \min(2,3) = 4$ KHz.

6. $x_1(2t) \longrightarrow$ Scaling $f_s = a(2f_m)$

$= 2(2 \times 2)$

$= \underline{8 \text{ KHz}}$.



Nyquist Sampling

7. $x_2(t/4) \longrightarrow f_s = 9(2f_2) = \frac{1}{4} \times 2 \times 3 = \frac{3}{2}$ KHz.

8. $\frac{dx_1(t)}{dt} \longrightarrow f_s = 2f_1 = 2 \times 2 = 4$ KHz

9. $x_2(t-7) \longrightarrow f_s = 2f_2 = 2 \times 3 = 6$ KHz.

Nyquist Sampling

10. $\left[x_2\left(\frac{t}{4}\right)\right]^2 = x_2\left(\frac{t}{4}\right) \cdot x_2\left(\frac{t}{4}\right)$

$\frac{3}{4}$ $\frac{3}{4}$

$\longrightarrow f_s = 2\left(\frac{3}{4}+\frac{3}{4}\right) = 3\ \text{KHz}$

11. $\frac{dx_1(t)}{dt} + \frac{dx_2(t)}{dt}$

$f_1 = 2$ $f_2 = 3$

$f_s = 2f_2 = \underline{6\ \text{KHz}}$.

12. $x_1(t-2) * x_2(-t+2)$

$f_s = 2 \times \min(f_1, f_2)$

$= 2 \times 2$

$= \underline{4\ \text{KHz}}$.

#Q. The Nyquist sampling interval, for the signal sinc(700t) + sinc(500t) is

multitone

$f_1 = 350\,Hz$ $\quad$ $f_2 = 250\,Hz$

$$T_s = \frac{1}{2f_1} = \frac{1}{2\times 350} = \frac{1}{700}\ sec$$

(A) 1/350 sec

(B) 1/700 sec ✓

(C) π/350 sec

(D) 1/300 sec

#Q. The spectrum of a bandlimited signal after sampling is shown below.

The value of sampling interval is



(oversampling)

f_m

$f_s - f_m = 150$

$f_s = 150 + f_m = 250\,Hz$

$T_s = \frac{1}{f_s} = \frac{1}{250} = 4ms$

A 1 msec

B 2 msec

C 4 msec ✓

D 8 msec

#Q. The Nyquist sampling rate, for the signal $s(t) = \frac{\sin(500\pi t)}{\pi t} \times \frac{\sin(700\pi t)}{\pi t}$ is given by

$f_1 = 250\,Hz$ $\quad$ $f_2 = 350\,Hz$

$f_s = 2(f_1 + f_2)$

$= 2(250 + 350)$

$= 1200\,Hz$

(A) 400 Hz

(B) 600 Hz

(C) 1200 Hz ✓

(D) 1400 Hz

$f_m = 1000\,Hz$.

$f_s = 1800\ Hz$.

#Q. A 1.0 kHz signal is sampled at a rate of 1800 samples per sec and the samples are applied to an ideal rectangular LPF with cut off frequency of 1100 Hz, then the output of the filter contains

(A) Only 800 Hz Component.

(B) 800 Hz and 900 Hz Components.

(C) 800 Hz and 1000 Hz Components. ✓

(D) 800 Hz, 900 Hz and 1000 Hz Components.

Freq. comp. present after sampling,

$f_m = 1000\,Hz$,

$f_s \pm f_m = 1800 \pm 1000 = 800\,Hz$ & $2800\,Hz$

$2f_s \pm f_m = 3600 \pm 1000 = 2600\,Hz$ & $4600\,Hz$

$f_s = 9\,kHz$ $f_m = 7\,kHz$

#Q. The signal $x(t) = \sin(14000\pi t)$, where t is in seconds, is sampled at a rate of 9000 samples per sec. The sample signal is the input to an ideal low pass filter with frequency response H(f) as follows

$M(f - nf_s)$

$$H(f) = \begin{cases} 1; & |f| \leq 12kHz \\ 0; & |f| > 12kHz \end{cases}$$

$f_c = 12\,kHz$

What is the number of sinusoids (N) in the output and their frequencies in KHz?

$f_m = 7\,kHz$

$f_s \pm f_m = 9 \pm 7 = 2\,kHz$ & $16\,kHz$

$2f_s \pm f_m = 18 \pm 7 = 11\,kHz$ & $25\,kHz$

A. N = 1, Frequency = 7

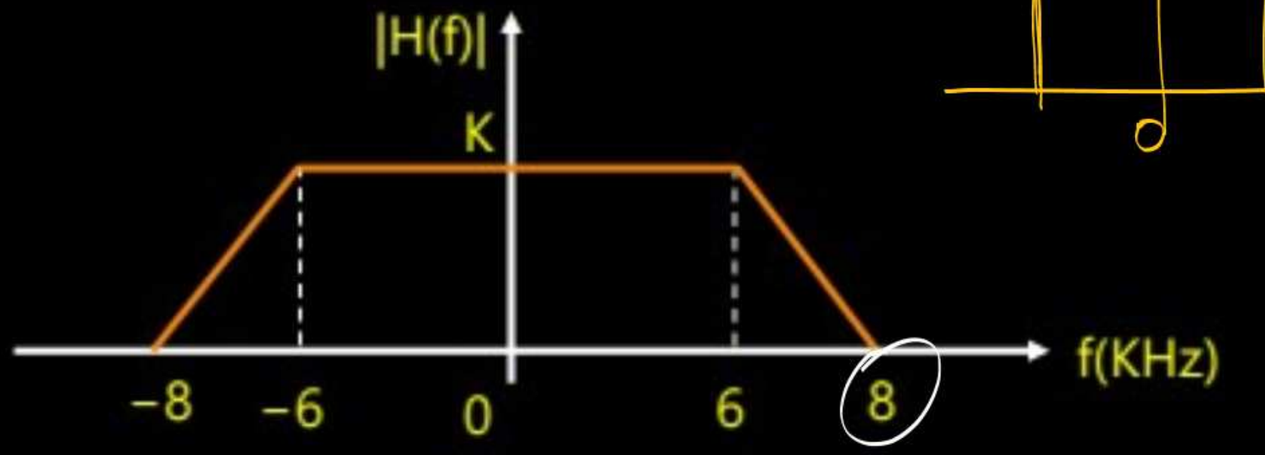
B. N = 3, Frequency = 2, 7, 11

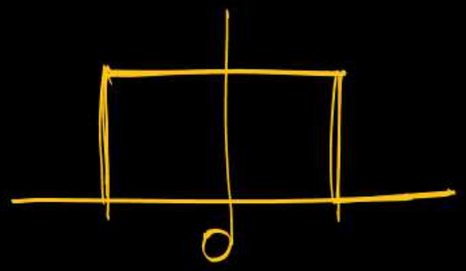
C. N = 2, Frequency = 2, 7

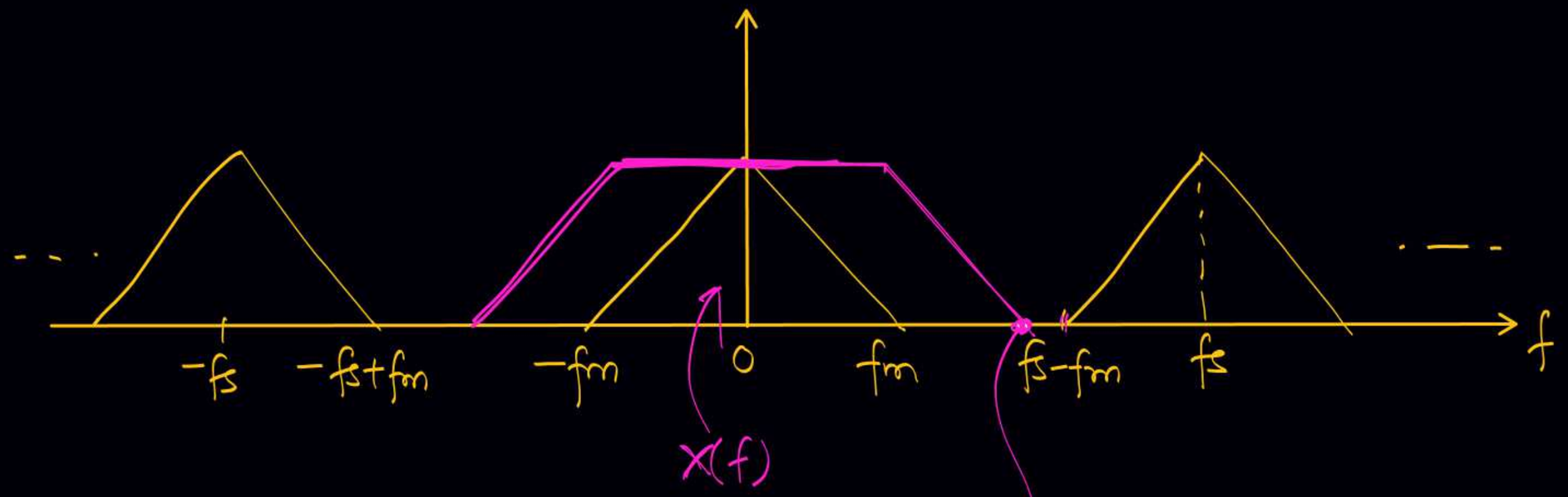
D. N = 2, Frequency = 7, 11

$f_m = 5KHz$

#Q. A bandlimited low pass signal x(t) of bandwidth 5 kHz is sampled at a sampling rate f_s. The signal x(t) is reconstructed using the reconstruction filter H(f) whose magnitude response is shown below.
The minimum sampling rate fs (in kHz) for perfect reconstruction of x(t) is 13







$8 < f_s - f_m$

$f_s > 8 + f_m$

$f_s > 8 + 5$

$f_s > 13\ \text{kHz}$

#Q. Consider a sampled signal $y(t) = 5 \times 10^{-6}\, x(t) \sum_{n=-\infty}^{\infty} \delta(t - nT_s)$ where $x(t) = 10 \cos(8\pi \times 10^3\, t)$ and $T_s = 100\ \mu sec$. When $y(t)$ is passed through an ideal low pass filter with a cut off frequency of 5 KHz, the output of the filter is

(A) $5 \times 10^{-6} \cos(8\pi \times 10^3 t)$

(B) $5 \times 10^{-5} \cos(8\pi \times 10^3 t)$

(C) $5 \times 10^{-1} \cos(8\pi \times 10^3 t)$

(D) $10 \cos(8\pi \times 10^3 t)$

#Q. Consider two real valued signals, x(t) bandlimited to [−500 Hz, 500 Hz] and y(t) bandlimited to [−1 KHz, 1KHz]. For z(t) = x(t).y(t), the Nyquist sampling frequency (in kHz) is 3 KHz

$f_1 = 500Hz, \quad f_2 = 1000Hz$

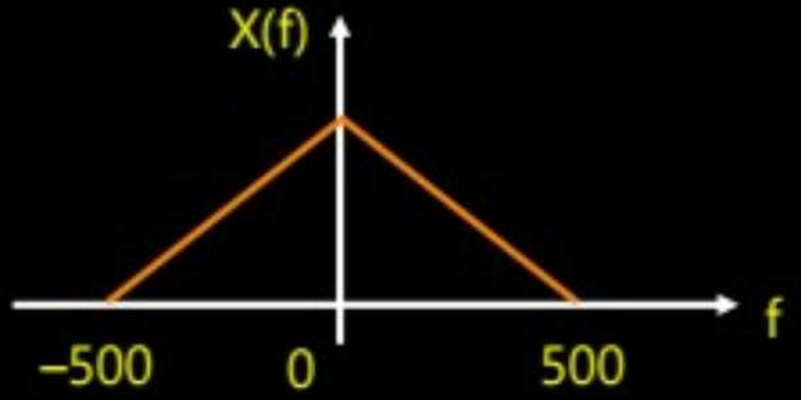
$f_s = 2(f_1 + f_2) = 3000$

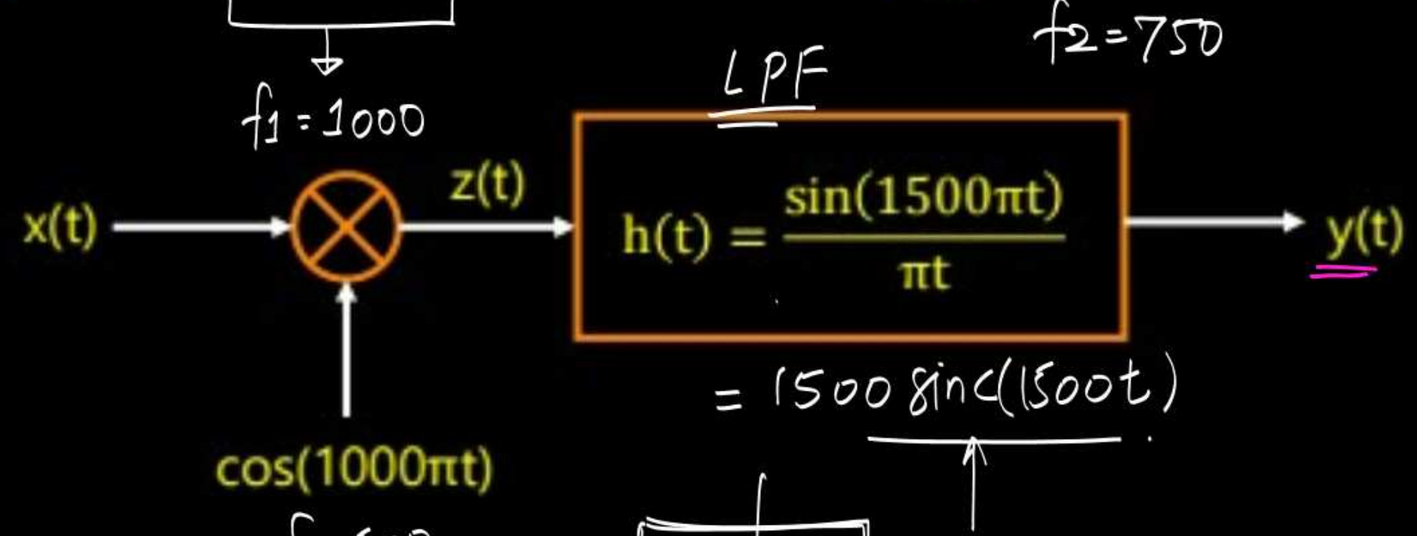
#Q. The output y(t) of the following system is to be sampled so as to reconstruct from its samples uniquely. The required minimum sampling rate for y(t) is _____

$y(t) = x(t)\cos(1000\pi t) * \frac{\sin(1500\pi t)}{\pi t}$

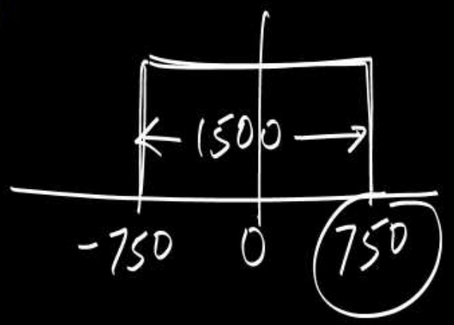
500 500

$f_1 = 1000$



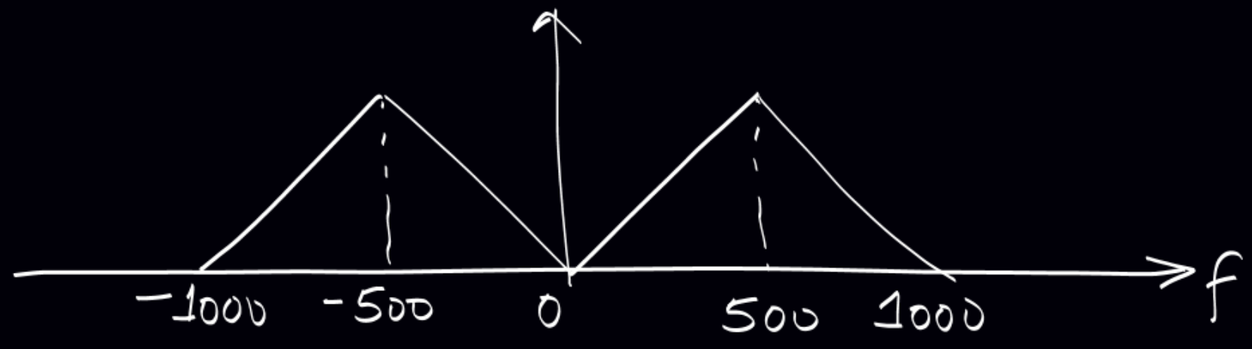


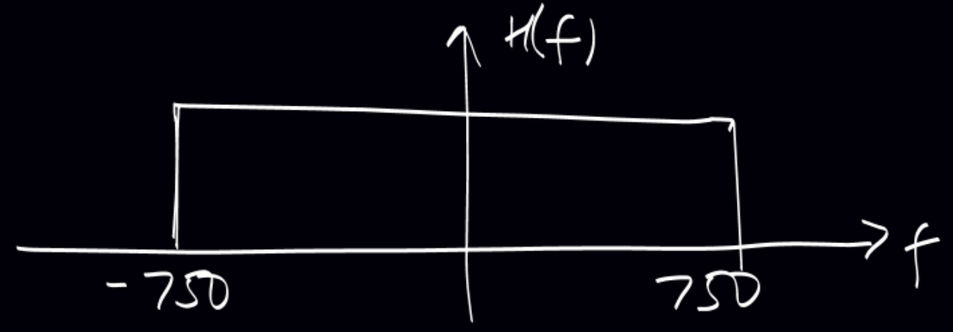
$= 1500\,\text{sinc}(1500t)$

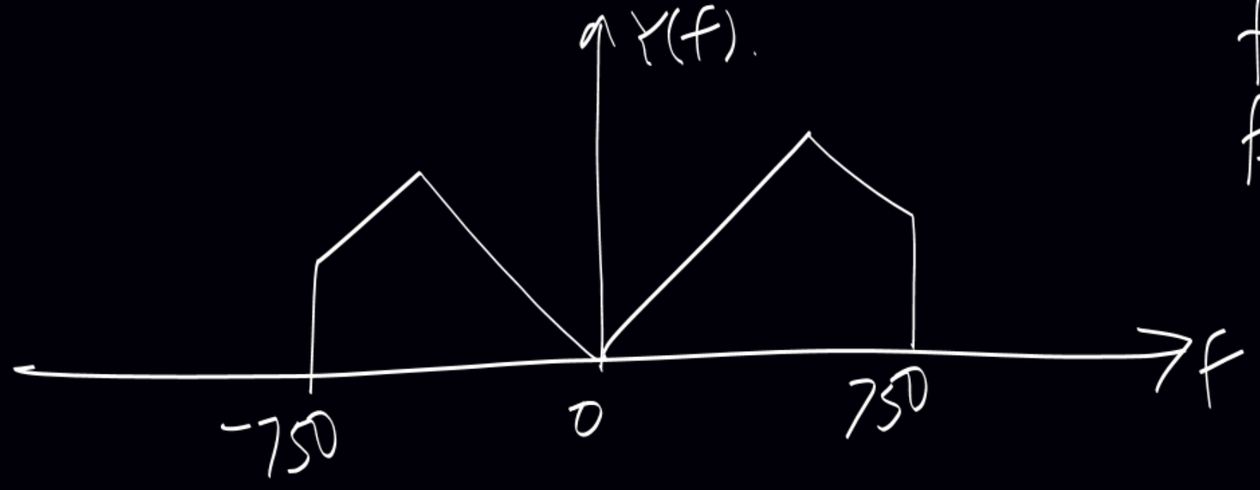


$f_s = 2 \times \text{min freq}$

$= 2 \times 750 = 1500\,\text{Hz}$



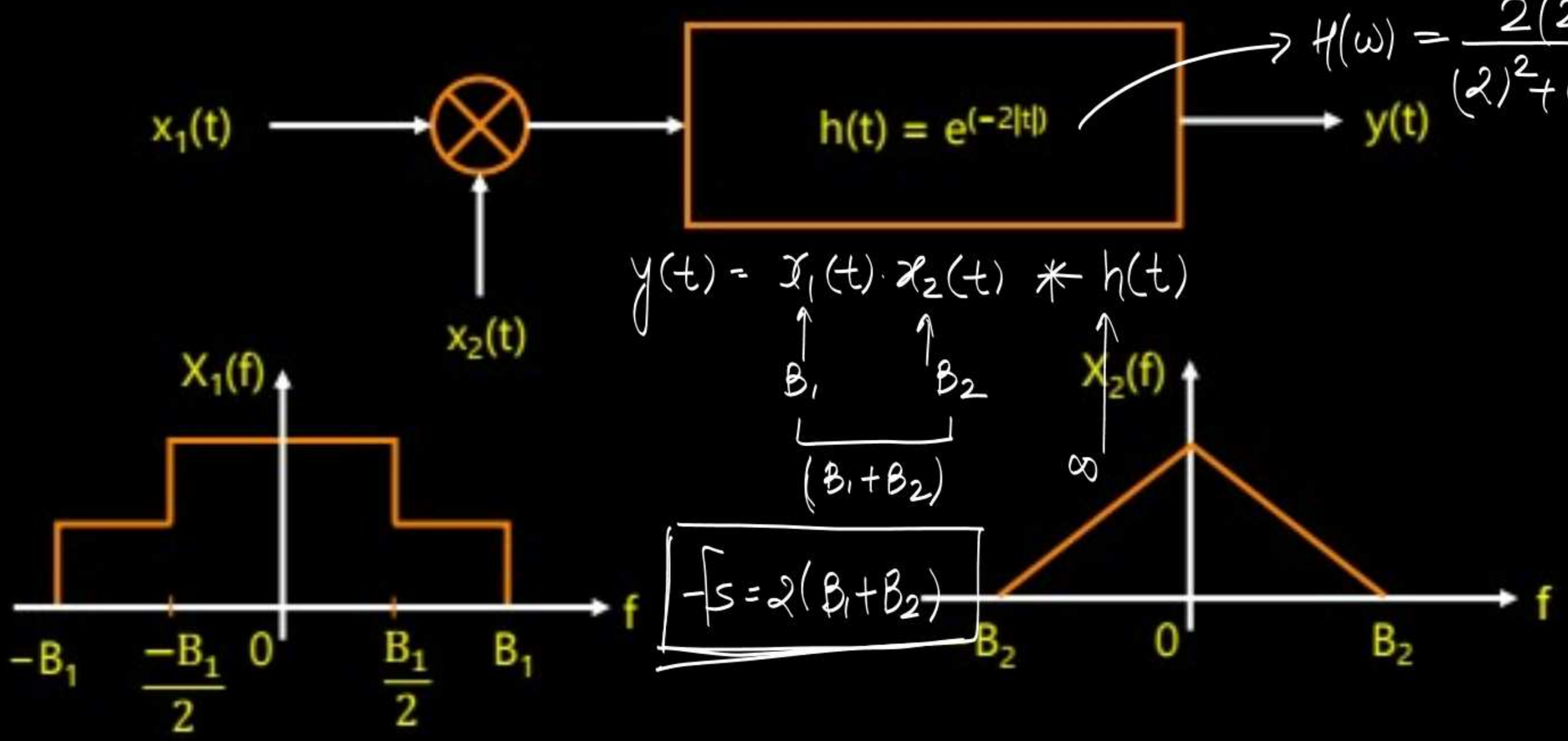




$f_m = 750\ Hz$

$f_s = 2f_m = 1500\ Hz$

#Q. Let Fourier transform of $x_1(t)$ is $X_1(f)$ and $x_2(t)$ is $X_2(f)$. For the system shown below the minimum sampling rate required to sample $y(t)$ so that $y(t)$ can be reconstructed uniquely from its samples is _____



#Q. A bandlimited signal with a maximum frequency of 5 KHz is to be sampled. According to the sampling theorem, the sampling frequency which is/are NOT valid

(A) 5 KHz ✓
(B) 12 KHz
(C) 2.5 KHz ✓
(D) 20 KHz

$f_s \geq 2 f_m$

$f_s \geq 2 \times 5$

$f_s \geq 10\ KHz$

#.Q Consider a real valued baseband signal x(t) bandlimited to 10 KHz.

The Nyquist rate for the signal $y(t) = x(t).x\left(1+\frac{t}{2}\right)$ is

(A) 60 KHz

(B) 15 KHz

(C) 20 KHz

(D) 30 KHz ✓

$f_1 = 10KHz$ $\quad f_2 = \frac{10}{2} = 5KHz$

$f_s = 2(f_1 + f_2) = 30KHz$

$f_1 = 5\,Hz$ $f_2 = 15\,Hz$

#Q. Let $x(t) = \cos(10\pi t) + \cos(30\pi t)$ be sampled at 20 Hz and reconstructed using an ideal low pass filter with cut off frequency of 20 Hz. The frequency/frequencies present in the reconstructed signal is/are

multitone

$f_1 = 5\,Hz,\ f_2 = 15\,Hz$

$f_s \pm f_1 = 20 \pm 5 = 15 \& 25$

$2f_s \pm f_1 = 40 \pm 5 = 35 \& 45$

$f_s \pm f_2 = 20 \pm 15 = 5 \& 35$

$2f_s \pm f_2 = 40 \pm 15 = 55 \& 25$

(A) 5 Hz and 15 Hz only.

(B) 10 Hz and 15 Hz only.

(C) 5 Hz,10 Hz and 15 Hz only.

(D) 5 Hz only.

THANK you